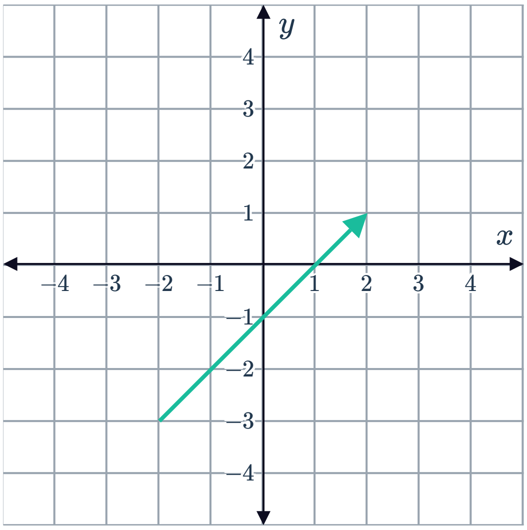
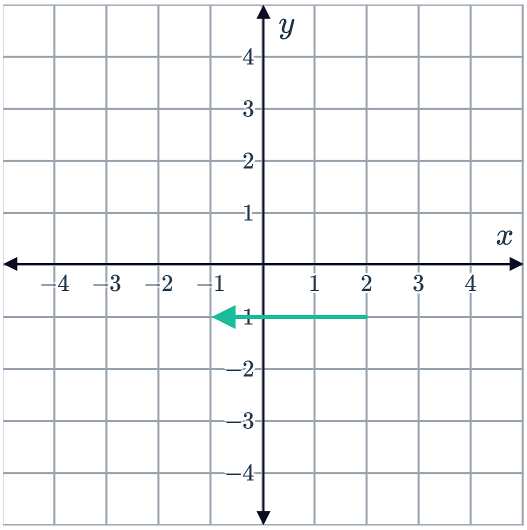
Column vectors

1 Write the vector represented on the following planes as a column vector:

a

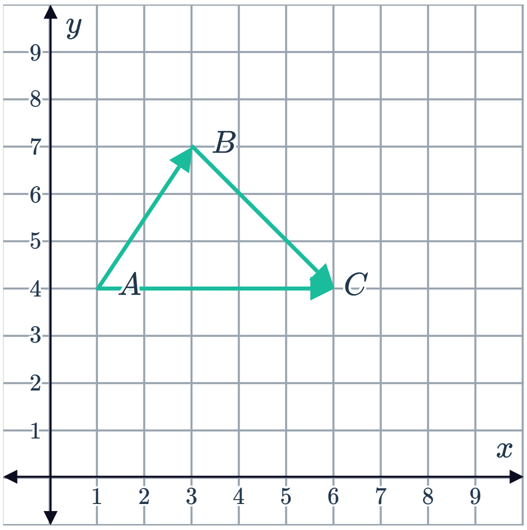


b



2 Consider the following figure and write the column vector for each of the following:

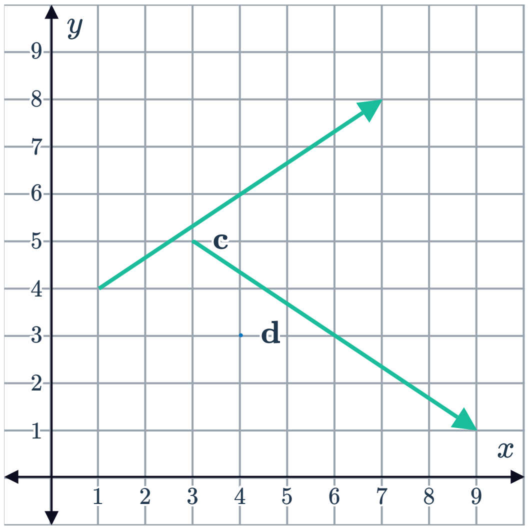
a $\overrightarrow{AB}$ **b** $\overrightarrow{BC}$ **c** $\overrightarrow{AC}$



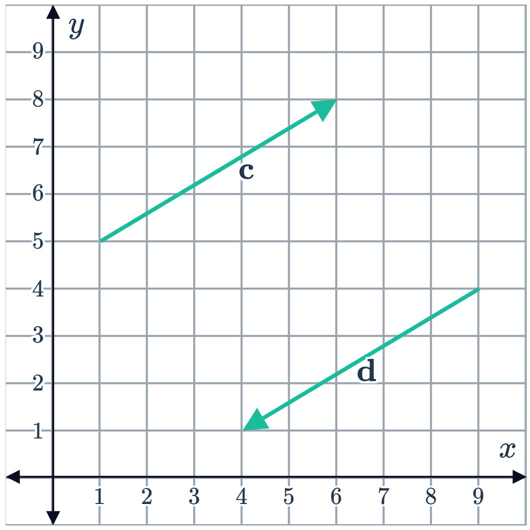
3 For each of the following graphs:

i Write the vector c as a column vector. **ii** Write the vector d as a column vector.

a



b



4 If $A = (-2, 3)$, $B = (1, 7)$, $C = (-9, 6)$ and O is the origin, express each of the following as a column vector:

a $\overrightarrow{OB}$ **b** $\overrightarrow{AC}$ **c** $\overrightarrow{AO}$ **d** $\overrightarrow{CB}$

5 Plot the vector $\begin{bmatrix} 5 \\ 9 \end{bmatrix}$ on a number plane. Use the origin as the starting point of the vector.

6 Let $A = (-6, 2)$, $B = (1, 4)$, $C = (-3, 5)$ and O be the origin. Plot the following vectors on a number plane:

a $\overrightarrow{OC}$ **b** $\overrightarrow{AB}$ **c** $\overrightarrow{BO}$ **d** $\overrightarrow{OC}$

7 For each of the following sets of points:

i Plot the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ on on a number plane.

ii State whether the two vectors are equivalent.

a $A\,(5, 4)$, $B\,(5, 9)$, $C\,(-4, -2)$ and $D\,(-4, 3)$.

b $A\,(-4, 2)$, $B\,(-6, 6)$, $C\,(-4, -2)$ and $D\,(-6, 2)$.

c $A\,(3, -4)$, $B\,(3, -6)$, $C\,(5, 3)$ and $D\,(5, 5)$.

8 State whether each of the following is a v[redacted] quantity:

a A force of $8\ \mathrm{N}$ acting horizontally right.

b A displacement of $4\ \mathrm{m}$ along the line joining A and B.

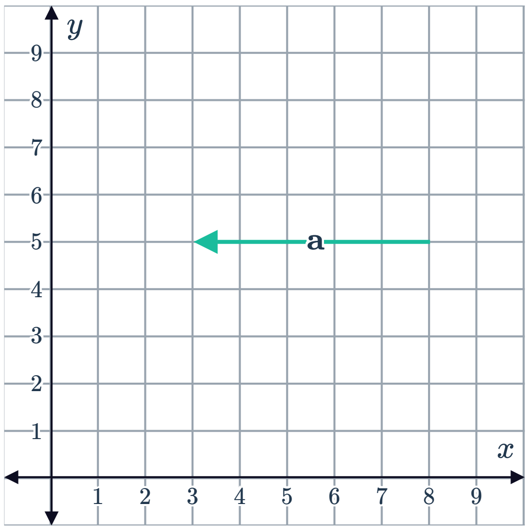
c A mass of $1\ \mathrm{kg}$

d A time of 1 second.

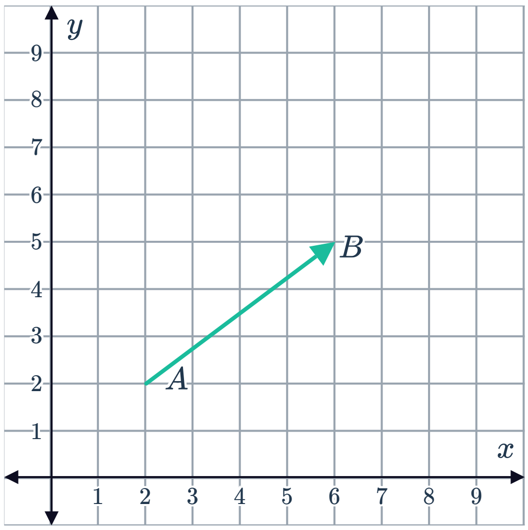
9 What is the opposite of the vector representing 800 km south?

Magnitude of a vector (Extended)

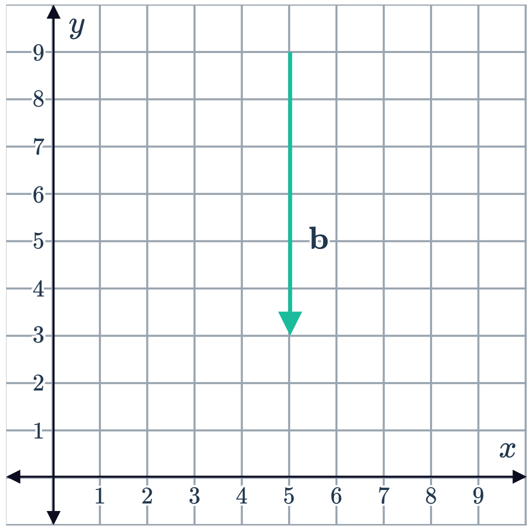
10 Find the magnitude of vector a:



11 Find the magnitude of the vector between points A and B:



12 Find the magnitude of vector b:

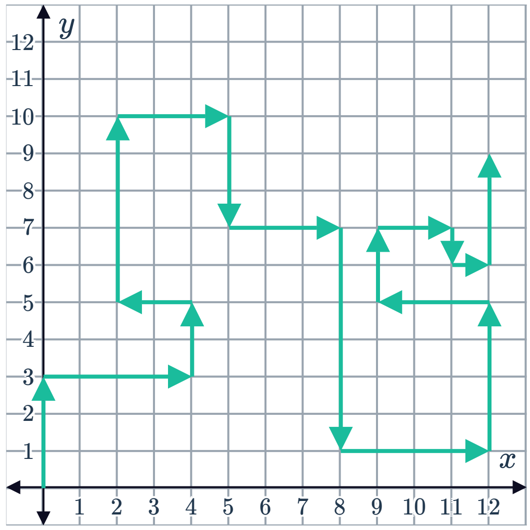


13 Consider the plotted path:

a Calculate the displacement of the path.

b Calculate the distance of the path.

c Is the displacement equal to the distance? Explain your answer.



14 Consider the vector defined by the directed line segment from $(-5, 3)$ to $(1, -5)$.

a Plot the vector.

b Find the magnitude of the vector.

15 Two vectors that are parallel can have different magnitudes. Is it true or false?

16 Find the exact magnitudes of the following vectors

a $\begin{bmatrix} -7 \\ 0 \end{bmatrix}$

b $\begin{bmatrix} 5 \\ -7 \end{bmatrix}$

c $\begin{bmatrix} 8 \\ 15 \end{bmatrix}$

d $\begin{bmatrix} -8 \\ -15 \end{bmatrix}$

e $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$

f $\begin{bmatrix} -6 \\ 0 \end{bmatrix}$

17 Find the exact magnitude of the vector $(4, 8)$.

18 Find the magnitude of the vector $(12, -4, 3)$.

19 If $a = (3x, 4x)$ and $|a| = 15$, find the value o[illegible]

20 A pilot flies 12 km north and then 16 km east. Find the magnitude of the vector created from her initial position to the final position.

21 A boat travels 12 km west and then 5 km south. Determine the distance of the boat from its initial position.